Supplementary Numerical Analyses for Minimizing Upper Confidence Bounds: A Data-Driven Framework for Stochastic Programming

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In this supplementary material, we report two additional experiments that complement the main paper. The goal is to test whether the main conclusions of APUB-M—improved small-sample robustness, an interpretable nominal level, and convergence toward SAA-type behavior as the sample size grows—continue to hold beyond the baseline setting.

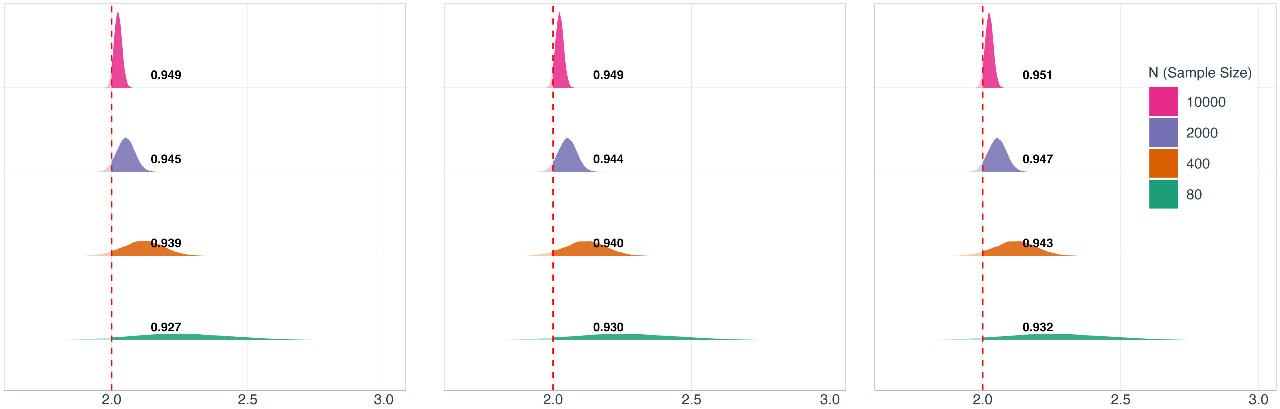
- Section 1 illustrates the asymptotic properties of APUB compared to the standard large-sample UCB and Efron's Percentile UCB.
- Section 2 compares the performance of APUB-M and Wasserstein DRO when used for a two-stage product-mix problem with fixed recourse;
- Section 3 conducts the sensitivity analysis of APUB-M on a multi-product newsvendor problem.

1 Comparison between the Standard Large-Sample UCB, Efron's Percentile UCB and APUB

Let $\xi \sim \text{Gamma}(2,1)$ and $F(\xi) = \xi$. So the population mean $\mu = 2$. We compare APUB with Efron's percentile-based upper bound and the standard large-sample upper bound given as $\widehat{\mu}_N + z_\alpha \widehat{\sigma}_N/\sqrt{N}$, where z_α denotes z critical value. In order to estimate the probability density functions (pdf) of three upper bounds, we performed a Monte Carlo simulation with $\alpha = 0.05$ while allowing the sample sizes, N, to vary from 80 to 10,000.

We make two observations:

- **APUB exhibits asymptotic correctness as an upper confidence bound.** As the sample size increases, the coverage probability of APUB remains above the nominal level, rather than converging exactly to it. By contrast, Efron's bound and the standard upper bound become closer to the nominal level. This highlights that APUB is asymptotically correct, but not asymptotically exact.
- **APUB is asymptotically consistent.** As the sample size increases, their sampling distributions become more concentrated around the true mean. In particular, APUB becomes increasingly tight and converges to the correct limit, consistent with the asymptotic consistency established in the main paper.



(a) Standard Upper Bound (b) Efron's Upper Bound (c) APUB

Figure 1: The comparison between APUB, Efron's upper bound, and the standard large-sample upper bound.

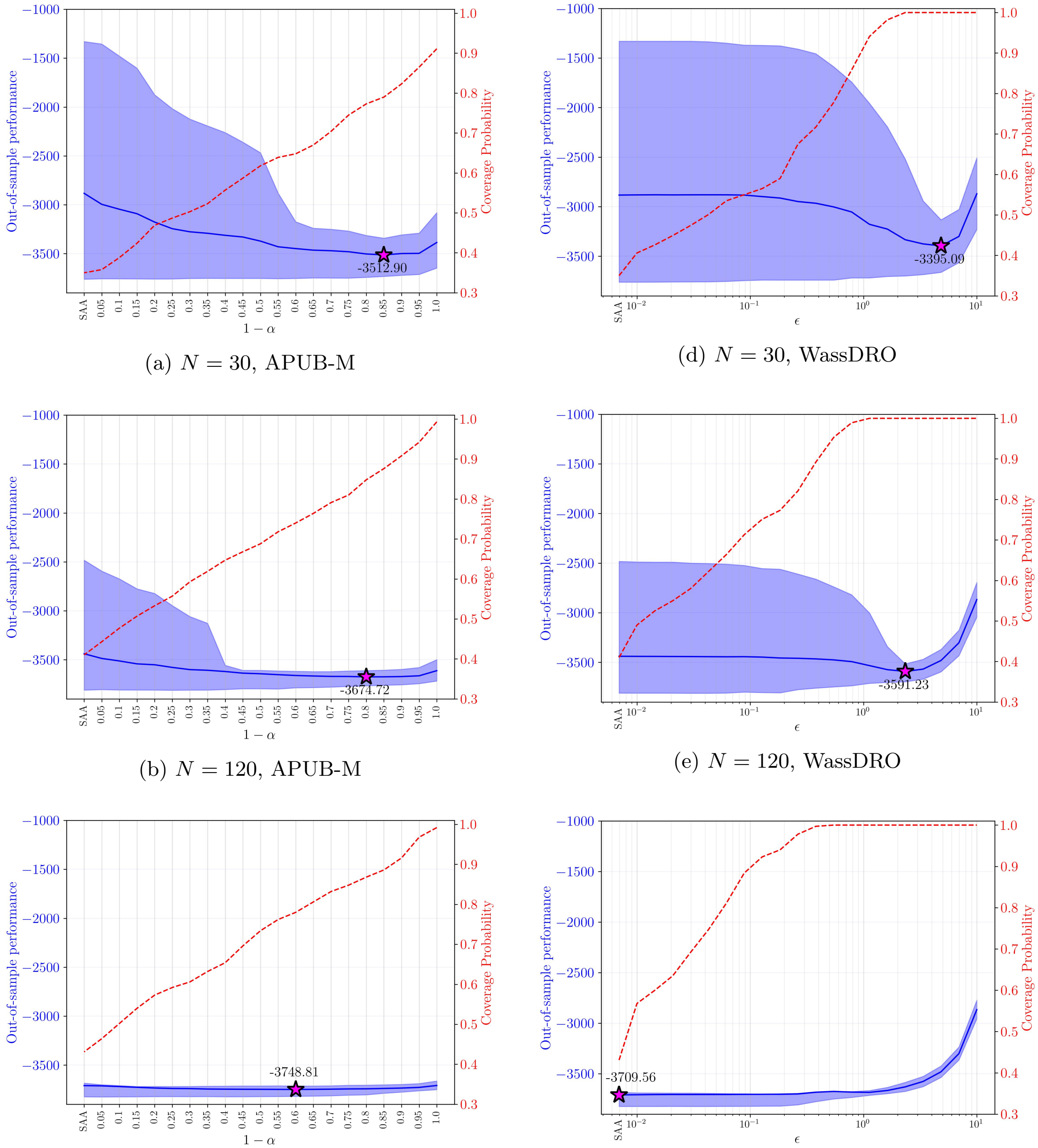
2 Comparison with Wasserstein DRO on a Two-Stage Fixed Recourse Problem

We compare APUB-M with Wasserstein DRO (WassDRO) using a two-stage product-mix problem with fixed recourse (King, 1988) (detailed parameters in Appendix A). Each method is trained on a sample of size N, and out-of-sample performance is evaluated on a large independent test set over 1,000 trials. Figure 2 illustrates the mean out-of-sample cost, the 10th–90th percentile performance band, and the empirical coverage probability. Note that the leftmost point in each APUB-M panel ($\alpha = 1$) corresponds to the standard SAA-M. We highlight the following key observations:

- **Statistical Interpretability vs. Geometric Tuning:** APUB-M utilizes a nominal confidence level $(1-\alpha)$ with a rigorous frequentist interpretation. In contrast, selecting a statistically meaningful radius ϵ for WassDRO remains a significant challenge, often requiring heuristic-based calibration. As shown in Figures 2a–2c, APUB-M demonstrates asymptotic correctness, where the empirical coverage probability (dashed line) consistently converges toward the nominal confidence level $(1 - \alpha)$ as N increases. This provides a transparent "uncertainty knob" that is missing in geometric DRO approaches.

- **Consistent Data-Driven Convergence:** APUB-M is inherently data-driven, achieving consistency for any fixed $(1 - \alpha)$ as the bootstrap distribution naturally contracts with N. Conversely, WassDRO relies on a radius ϵ that typically requires manual N-dependent scaling to prevent vanishing consistency or extreme over-conservatism. Figures 2a–2c illustrate that APUB-M automatically modulates its robustness: as epistemic uncertainty diminishes with larger N, the APUB-M objective naturally converges toward the true mean without manual parameter decay.

- **Controlled Conservatism and Reliability:** In low-data regimes ($N = 30$), "worst-case" scenarios are often under-represented. APUB-M explicitly accounts for the epistemic uncertainty of these tail events, ensuring reliability without the structural over-conservatism seen in WassDRO. As shown in Figure 2f, an inappropriate choice of ϵ in WassDRO leads to a sharp increase in mean cost (over-conservatism). APUB-M maintains a more stable performance pro-



(a) $N = 30$, APUB-M

(d) $N = 30$, WassDRO

(b) $N = 120$, APUB-M

(e) $N = 120$, WassDRO

(c) $N = 480$, APUB-M

(f) $N = 480$, WassDRO

Figure 2: Mean out-of-sample cost (solid line), 10th–90th percentile band (shaded area), and empirical coverage probability (dashed line) for APUB-M and WassDRO. Stars mark the minimum mean cost.

file by anchoring its robustness to the statistical variance of the mean rather than an arbitrary geometric ball.

3 Sensitivity Analysis of APUB-M on Multi-Product Newsvendor Problem

We evaluate the APUB-M framework using a 10-product newsvendor problem based on the formulation by Hanasusanto et al. (2015) (detailed parameters are provided in Appendix A.2). This benchmark is particularly relevant as Mohajerin Esfahani and Kuhn (2018) have noted that Wasserstein-based DRO may encounter performance limitations in settings like the newsvendor problem, where the random loss function possesses a Lipschitz modulus with respect to the random scenarios that is independent of the decision variables. To assess the robustness of APUB-M, we consider two distinct demand scenarios:

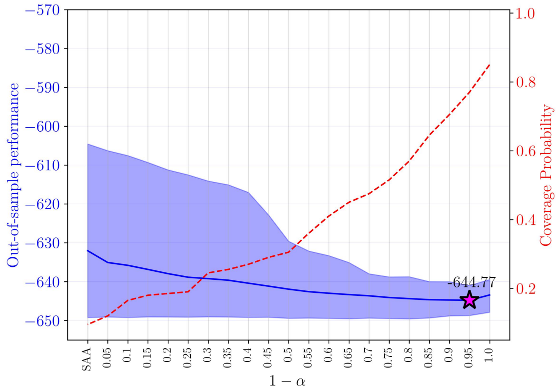
- **Case I:** Demand follows a standard Gaussian-mixture distribution, representing a multimodal but well-behaved environment.
- **Case II:** Demand is subjected to additional biased noise, creating higher variance and a more complex underlying true distribution to simulate a high-ambiguity environment.

3.1 Out-of-Sample Behavior

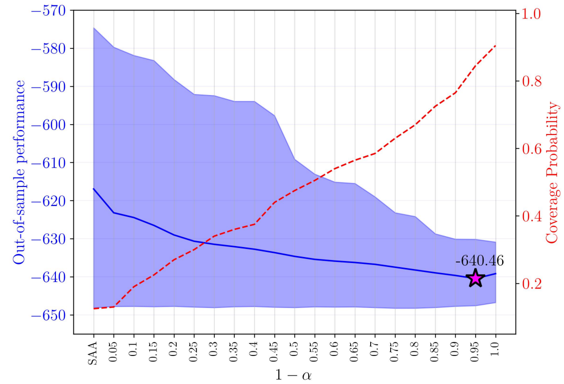
Figure 3 summarizes the out-of-sample behavior of APUB-M across two experimental settings. The primary trend remains consistent across all panels: APUB-M consistently outperforms SAA-M in both mean cost and performance variability. These gains are most pronounced when data are scarce ($N = 30$) or when the underlying distribution exhibits high ambiguity. Specifically, Case II represents a more challenging environment than Case I. At $N = 30$, the performance band in Case II is significantly wider, and the nominal confidence level that minimizes the mean cost is notably higher. This suggests that heightened ambiguity necessitates a more conservative APUB objective to maintain reliability. As the sample size increases from $N = 30$ to $N = 120$, the performance gap between the two cases narrows, demonstrating that APUB-M effectively adapts to decreasing epistemic uncertainty while continuing to leverage the information in larger datasets.

3.2 Stability of Optimal Solutions

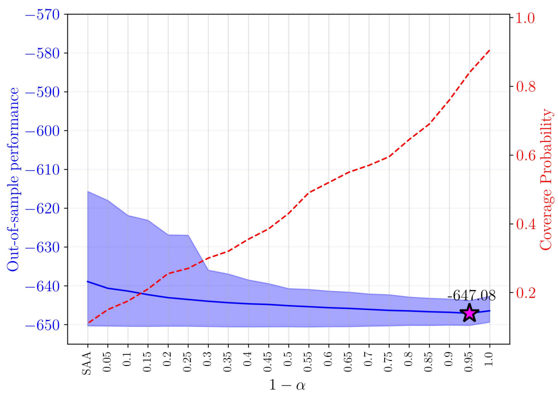
Figure 4 visualizes the resulting order quantities for Case I. When data are scarce ($N = 30$), the optimal solution is highly sensitive to the nominal level $(1 - \alpha)$, particularly for specific products (e.g., P2). As N increases, these sensitivity curves flatten, and the APUB-M solutions smoothly converge toward the SAA-M estimates. This behavior mirrors the performance trends and reinforces the theoretical consistency of the framework.A critical advantage of APUB-M is the prescriptive stability of its solutions across varying sample sizes. To quantify this, we measure the change in the recommended order vector from $N = 30$ to $N = 120$ using the ℓ_2-norm. For SAA-M, the shift is 7.89. In contrast, APUB-M yields significantly more stable decisions, with a shift of only 3.99 at $(1 - \alpha) = 0.5$ and 3.36 at $(1 - \alpha) = 0.95$. These results indicate that APUB-M provides materially more reliable and consistent prescriptions in data-poor environments, reducing the nervousness of the optimization results as more data are collected.



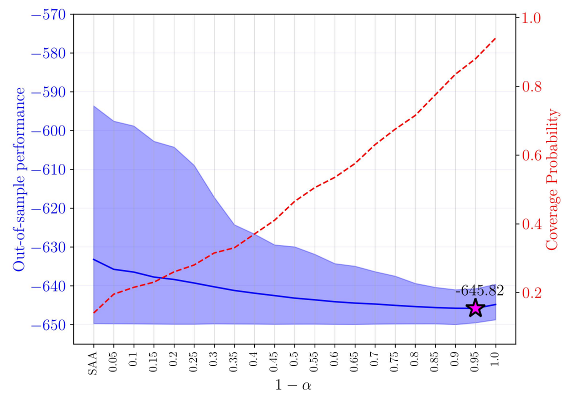
(a) Case I, $N = 30$



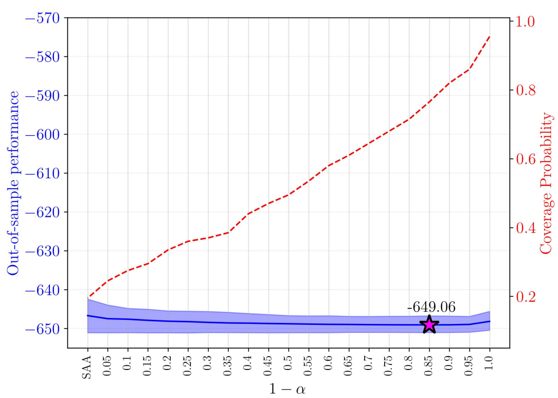
(b) Case II, $N = 30$



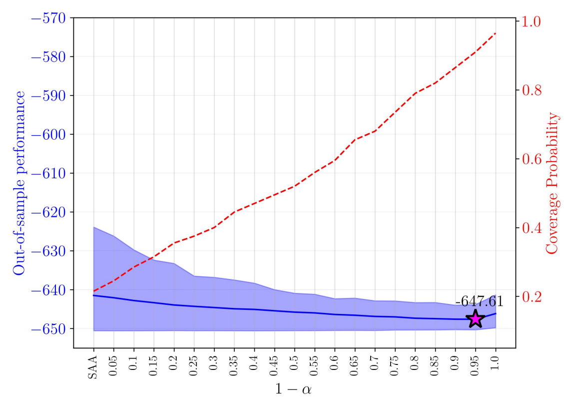
(b) Case I, $N = 60$



(e) Case II, $N = 60$

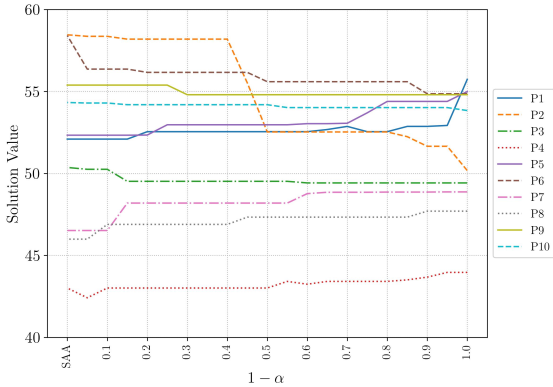


(c) Case I, $N = 120$

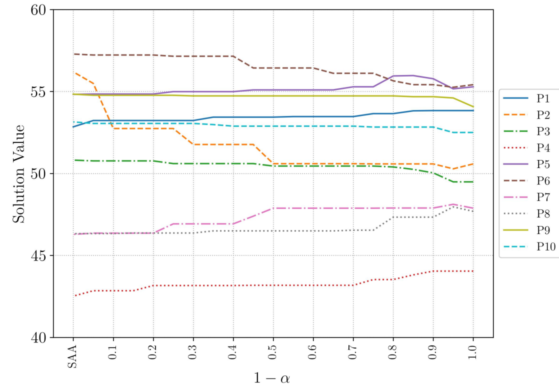


(f) Case II, $N = 120$

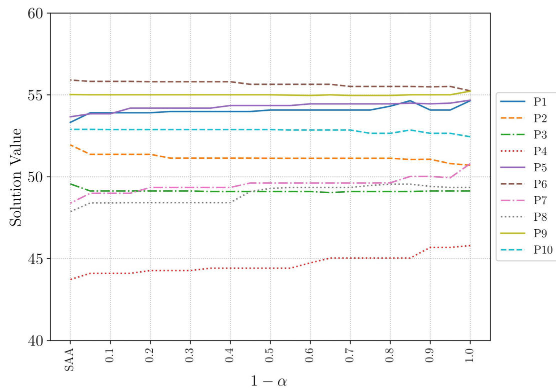
Figure 3: Mean out-of-sample cost (solid line), 10th–90th percentile band (shaded area), and empirical coverage probability (dashed line) for APUB-M under two ambiguity levels. Stars mark the minimum mean cost.



(a) $N = 30$



(b) $N = 60$



(c) $N = 120$

Figure 4: Optimal order quantities for the 10 products.

References

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A. J. King. Stochastic programming problems: Examples from the literature. *Numerical techniques for stochastic optimization*, pages 543–567, 1988.

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A Experimental Parameters

A.1 Parameters for the fixed-recourse product-mix problem

The two-stage fixed-recourse experiment in Section 2 uses

$$A = 0, \quad b = 0, \quad c = [-12, -20, -18, -40]^\intercal,$$
$$q = [6, 12, 0, 0]^\intercal, \quad h = [500\gamma_1, 500\gamma_2]^\intercal,$$
$$T = \begin{bmatrix} 4 - \frac{\gamma_1}{4} & 9 - \frac{\gamma_1}{4} & 7 - \frac{\gamma_1}{4} & 10 - \frac{\gamma_1}{4} \\ 3 - \frac{\gamma_2}{4} & 1 - \frac{\gamma_2}{4} & 3 - \frac{\gamma_2}{4} & 6 - \frac{\gamma_2}{4} \end{bmatrix}, \qquad W = \begin{bmatrix} -0.9 & 0 & 1 & 0 \\ 0 & -0.9 & 0 & 1 \end{bmatrix},$$

where

$$[\gamma_1, \gamma_2]^\intercal \sim \frac{7}{10}\mathcal{N}\left(\begin{bmatrix} 12 \\ 8 \end{bmatrix}, \begin{bmatrix} 5.76 & 1.92 \\ 1.92 & 2.56 \end{bmatrix}\right) + \frac{3}{10}\mathcal{N}\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0.16 & 0.04 \\ 0.04 & 0.04 \end{bmatrix}\right).$$

A.2 Parameters for the newsvendor problem

The newsvendor experiment in Section 3 uses

$$F(x, \xi) = p^\intercal x + h^\intercal (x - \xi)_+ + b^\intercal (\xi - x)_+.$$

Here x is the order vector, ξ is the random demand vector, and p, h, and b are the unit purchase, overage, and underage costs.

$$p = -2, \qquad h = 9, \qquad b = 5.$$

$$\mu_1 = [60.89, 48.58, 46.81, 56.54, 61.58, 52.69, 69.42, 60.54, 54.43, 51.76]^\intercal$$

$$\mu_2 = [50.30, 61.87, 53.16, 41.79, 51.94, 62.14, 45.47, 45.26, 55.95, 55.95]^\intercal$$

$$\Sigma_1 = \begin{bmatrix}
9.27 & 2.84 & -0.07 & 1.19 & -0.48 & 1.40 & 2.87 & 4.06 & -1.40 & -1.96 \\
2.84 & 5.90 & -2.83 & 0.21 & 2.27 & -2.40 & -0.89 & 4.22 & 3.43 & 2.78 \\
-0.07 & -2.83 & 5.48 & -0.30 & 0.90 & 3.54 & -4.51 & -2.45 & -2.91 & -4.95 \\
1.19 & 0.21 & -0.30 & 7.99 & -1.02 & -1.27 & -0.15 & -1.55 & -1.69 & -0.36 \\
-0.48 & 2.27 & 0.90 & -1.02 & 9.48 & -0.08 & -3.69 & 2.71 & -0.69 & -0.34 \\
1.40 & -2.40 & 3.54 & -1.27 & -0.08 & 6.94 & -1.26 & -2.73 & 0.01 & -5.19 \\
2.87 & -0.89 & -4.51 & -0.15 & -3.69 & -1.26 & 12.05 & -0.16 & -0.16 & 2.44 \\
4.06 & 4.22 & -2.45 & -1.55 & 2.71 & -2.73 & -0.16 & 9.16 & -0.77 & 1.94 \\
-1.40 & 3.43 & -2.91 & -1.69 & -0.69 & 0.01 & -0.16 & -0.77 & 7.41 & 2.24 \\
-1.96 & 2.78 & -4.95 & -0.36 & -0.34 & -5.19 & 2.44 & 1.94 & 2.24 & 6.70
\end{bmatrix}$$

$$\Sigma_2 = \begin{bmatrix}
6.32 & 2.99 & -0.06 & 0.73 & -0.33 & 1.36 & 1.55 & 2.51 & -1.19 & -1.75 \\
2.99 & 9.57 & -4.09 & 0.19 & 2.44 & -3.60 & -0.74 & 4.02 & 4.49 & 3.83 \\
-0.06 & -4.09 & 7.06 & -0.25 & 0.86 & 4.74 & -3.35 & -2.08 & -3.40 & -6.08 \\
0.73 & 0.19 & -0.25 & 4.37 & -0.64 & -1.11 & -0.07 & -0.86 & -1.29 & -0.29 \\
-0.33 & 2.44 & 0.86 & -0.64 & 6.74 & -0.08 & -2.04 & 1.71 & -0.60 & -0.31 \\
1.36 & -3.60 & 4.74 & -1.11 & -0.08 & 9.65 & -0.98 & -2.41 & 0.01 & -6.62 \\
1.55 & -0.74 & -3.35 & -0.07 & -2.04 & -0.98 & 5.17 & -0.08 & -0.10 & 1.72 \\
2.51 & 4.02 & -2.08 & -0.86 & 1.71 & -2.41 & -0.08 & 5.12 & -0.59 & 1.57 \\
-1.19 & 4.49 & -3.40 & -1.29 & -0.60 & 0.01 & -0.10 & -0.59 & 7.83 & 2.49 \\
-1.75 & 3.83 & -6.08 & -0.29 & -0.31 & -6.62 & 1.72 & 1.57 & 2.49 & 7.83
\end{bmatrix}$$

$$\varepsilon \sim \begin{bmatrix} \mathcal{U}(-5.37, 26.27) \\ \mathcal{U}(6.74, 14.16) \\ \mathcal{U}(3.22, 17.68) \\ \mathcal{U}(-7.48, 28.38) \\ \mathcal{U}(-4.89, 25.79) \\ \mathcal{U}(-0.21, 16.11) \\ \mathcal{U}(-12.14, 32.99) \\ \mathcal{U}(-7.74, 28.64) \\ \mathcal{U}(0.77, 20.13) \\ \mathcal{U}(2.13, 18.77) \end{bmatrix}.$$